

Original Article: Psychological Care

Self-monitoring of blood glucose changed non-insulin-treated Type 2 diabetes patients' beliefs about diabetes and self-monitoring in a randomized trial

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Accepted 8 August 2008

Abstract

Aims To determine whether differences in beliefs about diabetes and its treatment resulted from different intensities of self-monitoring of blood glucose (SMBG) in non-insulin treated patients with Type 2 diabetes in the Diabetes Glycaemic Education and Monitoring (DiGEM) trial.

Methods Patients ($n = 453$) were randomized to usual care, less-intensive SMBG and more intensive SMBG. Beliefs about diabetes were measured with a standard questionnaire (the revised Illness Perceptions Questionnaire; IPQ-R). Changes in beliefs were analysed using analysis of covariance (ANCOVA) with adjustment for baseline values. Mediation analyses assessed whether differences in behavioural outcomes between groups could be attributed to differences in beliefs.

Results Completed questionnaires were returned by 339 patients (74.8%). Respondents were mean ($\pm$ SD) age 65.9 $\pm$ 10 years and with diabetes duration of 4.8 $\pm$ 4.7 years (median 36, range 1–384 months). Concerns about the consequences of diabetes increased in both self-monitoring groups, relative to control subjects [$P = 0.004$; Cohen's d standardized effect size = 0.19 less intensive and $d = 0.36$ more intensive monitoring]. No other beliefs about diabetes differed between groups. Beliefs about the importance of self-testing increased in both self-monitoring groups relative to the usual-care group ($P < 0.001$; $d = 0.57$ less intensive and $d = 0.63$ more intensive monitoring). Changes in psychological well-being did not differ between groups, but control patients reported greater increases in general ($P = 0.014$) and specific ($P < 0.001$) dietary adherence than did patients in the self-monitoring groups. These outcomes were not mediated by intervention-related changes in beliefs.

Conclusions Despite changes in some beliefs about diabetes differing between groups there were no corresponding changes in self-reported health behaviours. This suggests that changes in illness beliefs resulting from SMBG do not cause changes in diabetes-related health behaviours.

Diabet. Med. 25, 1218–1228 (2008)

Keywords health beliefs, illness beliefs, illness perceptions, self-monitoring of blood glucose, Type 2 diabetes

Abbreviations BMQ, Beliefs about Medicines Questionnaire; CSM, Common Sense Model; DiGEM, Diabetes Glycaemic Education and Monitoring; HbA_{1c}, glycated haemoglobin; IPQ-R, revised Illness Perception Questionnaire; SDSCA, Summary of Diabetes Self-Care Activities; SMBG, self-monitoring of blood glucose; WBQ, Well-Being Questionnaire

Introduction

Self-monitoring of blood glucose is recommended as a means to improve glycaemic control for non-insulin-treated patients with Type 2 diabetes [1,2]. The rationale for self-monitoring in this group is that the procedure improves understanding of

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the impact of eating and physical activity on blood glucose levels [3,4], leading to greater understanding, improved self-efficacy and increased behavioural motivation. Previous studies have identified the potential of the approach to both motivate and discourage patients [5,6]. However, very few studies have examined the potential moderators or mediators of self-monitoring of blood glucose on glycated haemoglobin (HbA_{1c}) or self-care behaviours [7].

In-depth interviews with patients using self-monitoring of blood glucose have identified the importance of embedding self-monitoring within a comprehensive educational programme [8]. Such a programme needs to address interpretation of blood glucose readings, linkage of behaviour to blood glucose levels and implementation of plans for change in health-related behaviour [7]. However, there is also the potential for results of self-monitoring to lead to anxiety, through creating awareness of the limited impact that behaviours have over short-term fluctuations in blood glucose readings [5].

The Diabetes Glycaemic Education and Monitoring (DiGEM) open randomized controlled trial has suggested the glycaemic benefit of self-monitoring of blood glucose for non-insulin-treated Type 2 diabetes may be small and that persistence with more intensive self-monitoring was lower than that with less intensive self-monitoring [9]. In addition to testing the efficacy of self-monitoring, the trial was designed to explore its impact on beliefs about diabetes and self-reported health behaviour. The design of this aspect of the study was underpinned by psychological theory, specifically Leventhal's Common Sense Model (CSM) [10], which suggests that self-monitoring would lead to a stronger sense of personal and treatment control and a stronger illness identity, which in turn would motivate improved self-care activities [11]. We have therefore undertaken an analysis to examine whether, as predicted by the CSM, changes in beliefs about diabetes and beliefs about self-monitoring of blood glucose (SMBG) result in changes in self-reported diet, exercise and well-being and differ between patients exposed to SMBG over the course of 12 months and whether changes in self-reported behaviour are mediated through beliefs.

Methods

The DiGEM study was a 4-year, open, randomized, three-arm parallel trial. A full description of its design and conduct is given elsewhere [9]. The protocol was approved by the Oxfordshire B Research Ethics Committee.

The primary objective of the trial was to determine whether HbA_{1c} at 12 months was significantly different between patients with non-insulin-treated Type 2 diabetes receiving one of three allocated interventions: (i) standardized usual care with 3-monthly measurements of HbA_{1c} (control group); (ii) use of a meter with training focused on clinician interpretation of results (less intensive self-monitoring); (iii) use of a meter with training in self-interpretation and application of the results to diet, physical activity and medication adherence (more

intensive self-monitoring). We report here the analysis of measures of beliefs about diabetes, beliefs about self-monitoring of blood glucose and self-reports of self-care behaviours.

Patients and setting

The trial recruited patients sequentially from general practices in Oxfordshire and South Yorkshire. Between January 2003 and December 2005, 453 patients from 48 practices were randomized (Fig. 1) to receive usual care ($n = 152$), less intensive monitoring ($n = 150$) or more intensive self-monitoring ($n = 151$). Patients were eligible for randomization if they had Type 2 diabetes, were aged 25 years or more at diagnosis, were managed with diet or oral glucose-lowering agents alone, their HbA_{1c} at the assessment visit was $\geq 6.2\%$, they were independent in activities of daily living, they had not used a blood glucose meter twice a week or more over the previous 3 months and had no serious disease or limited life expectancy that would make intensive glycaemic control inappropriate.

Intervention

The CSM was used to guide development of the intervention and identify targets for measurement, with a specific causal model linking beliefs and behaviours (Fig. 2) [12]. Goal-setting techniques were used to facilitate behavioural change for all of the allocated interventions [13]. Patients allocated to the more intensive monitoring group were trained in interpretation of results and were encouraged to use SMBG to investigate the effect of their actions, such as checking levels after exercise or before and after meals. Participants were asked to view persistently elevated levels as a prompt to set new goals for behaviour change. Patients experiencing symptoms attributable to hypoglycaemia were encouraged to check their blood glucose to establish whether their symptoms were indeed as a result of low blood glucose [11].

An ongoing and iterative process designed to ensure that interventions were delivered as intended was carried out throughout the trial. All intervention visits were tape-recorded. A group consisting of a sociologist, clinical psychologist and members of the research team listened to over 150 of these tapes, with different people delivering the intervention and different stages of the intervention process. Any deviations from the protocol were fed back to the nurses delivering the interventions, as part of ongoing training. There did not appear to be any major or systematic deviations of intervention delivery from the specifications of the protocol. The process described constitutes better practice in terms of assessing fidelity of intervention delivery than is usual in studies of behavioural interventions [14].

Measures

A series of self-completed measures were used to assess patients' beliefs at the beginning of the trial and at the 12-month end-point (Table 1). The revised Illness Perceptions Questionnaire (IPQ-R) measures beliefs about a disease in eight domains [15]. Scales exploring beliefs about self-monitoring were developed

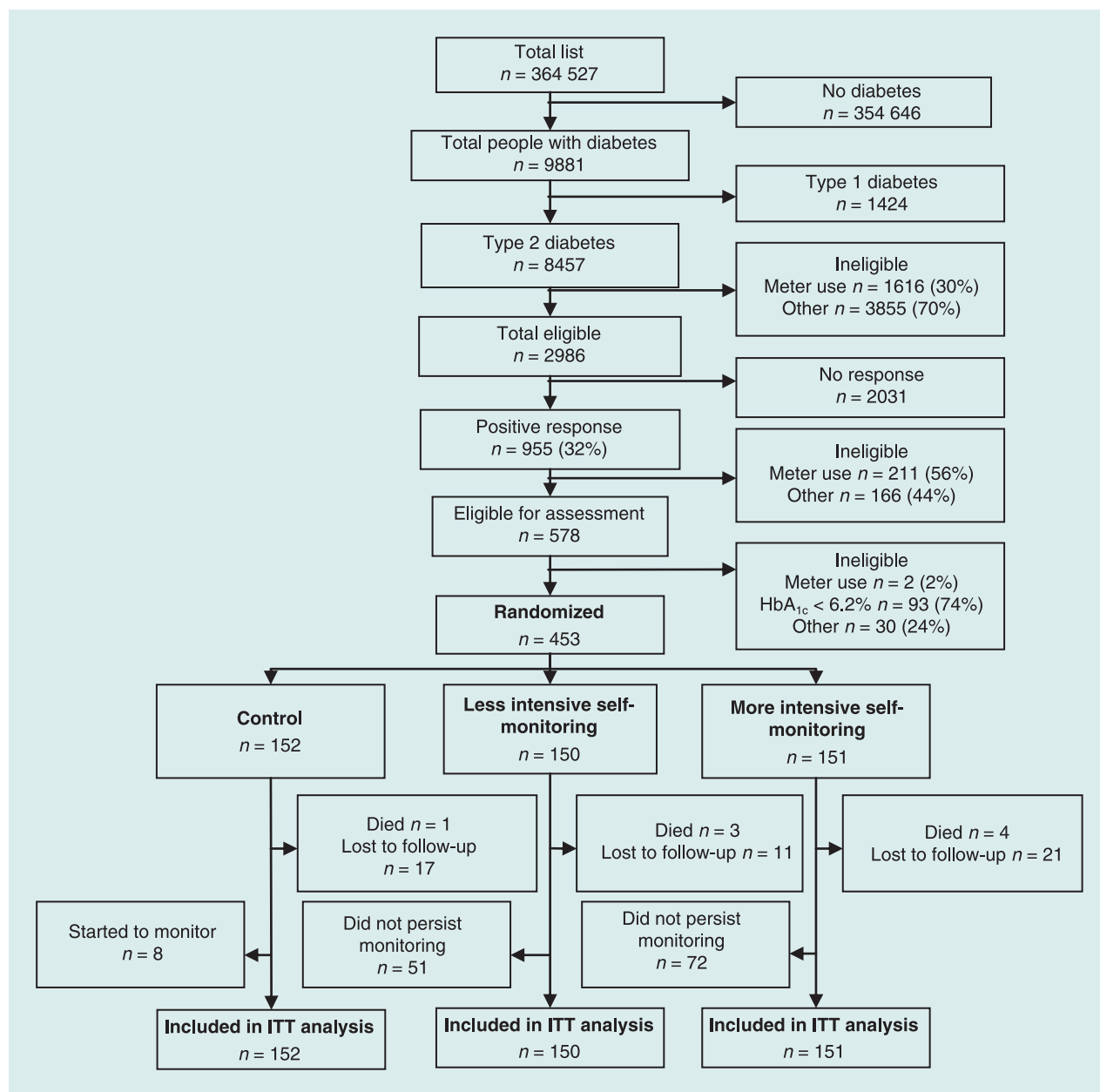


FIGURE 1 Trial disposition diagram. HbA_{1c}, glycated haemoglobin; ITT, intention to treat.

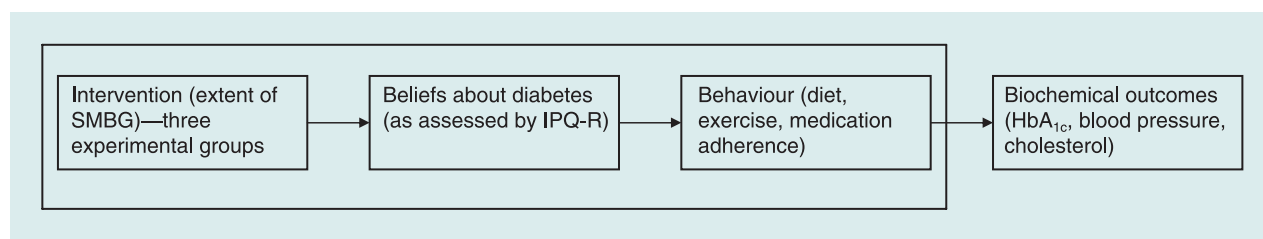


FIGURE 2 Hypothesized causal model linking beliefs, behaviour and outcomes (adapted from Leventhal, Nerenz & Steele, 1984). Note that this present paper only investigates the causal relationships depicted inside the box. HbA_{1c}, glycated haemoglobin; IPQ-R, revised Illness Perceptions Questionnaire; SMBG, self-monitoring of blood glucose.

Table 1 Measures and their characteristics

Scale	Variable name	Number of items	Cronbach's alpha	Sample question
Revised Illness Perception Questionnaire (IPQ-R) [15]	Identity	14	n/a	... experience of nausea
	Timeline—acute/chronic	6	$\alpha = 0.81$	... will last a long time
	Timeline—cyclical	4	$\alpha = 0.87$	... symptoms come and go
	Consequences	6	$\alpha = 0.68$	... has a major consequence on my life
	Personal control	6	$\alpha = 0.76$	... the course ... depends on me
	Treatment control	5	$\alpha = 0.40$	... my treatment will be effective in curing ...
	Illness coherence	5	$\alpha = 0.88$	... my diabetes is a mystery to me (R)
Self-monitoring blood glucose beliefs [16]	Emotional representations	6	$\alpha = 0.87$	... I get depressed when I think about ...
	Feel negative about self-testing	2	$\alpha = 0.65$	... checking at home would cause me to worry
	Symptoms make test unnecessary	2	$\alpha = 0.69$	... can tell my blood sugar is abnormal without testing it
	Important to self-test	2	$\alpha = 0.36$	... don't need to check more than once a week (R)
	Checking is painful	1		... checking my blood sugar is painful
Beliefs about Medicines Questionnaire (BMQ) [17]	Necessity	5	$\alpha = 0.82$	... my health depends on my medicines
Diabetes Self-Care Activities (SDSCA) [18]	Concerns	5	$\alpha = 0.77$	... I worry about the effects of these medicines
	General diet	2	$\alpha = 0.93$	... followed a healthy eating plan
	Specific diet	2	$\alpha = 0.08$	
	Specific diet: fruit and vegetables	1		... eat fruit and vegetables
	Specific diet: high-fat foods	1		... eat high-fat foods
	Exercise	2	$\alpha = 0.75$	... participate in at least 30 min of activity
Medication Adherence Report Scale (MARS) [19]	Self-report adherence scale	5	$\alpha = 0.62$	... forget to take my medicines (R)
Diabetes Treatment Satisfaction Questionnaire (DTSQ) [20]	Total satisfaction	6	$\alpha = 0.84$	... satisfied with current treatment
	Perceived frequency of hyperglycaemia	1		... blood sugars unacceptably high
	Perceived frequency of hypoglycaemia	1		... blood sugars unacceptably low
Well-Being Questionnaire (WBQ-12) [21]	Total well-being	12	$\alpha = 0.87$	
	Negative well-being	4	$\alpha = 0.83$	... get upset easily or feel panicky
	Energy	4	$\alpha = 0.75$	... been waking up feeling fresh and rested
	Positive well-being	4	$\alpha = 0.88$	... have lived the kind of life I wanted to

*Cronbach alpha coefficients based on questionnaires completed at baseline for the DiGEM trial.
(R), reverse scored item.

Table 2 Comparison of patients responding and not responding to questionnaires at 12-month follow-up

		Responding <i>n</i> = 361	Not responding <i>n</i> = 92	<i>P</i> value for difference*
Randomization group	Usual care	118 (32.7%)	34 (37.0%)	0.105
	Self-testing	128 (35.5%)	22 (23.9%)	
	Self-monitoring	115 (31.9%)	36 (39.1%)	
Age (years)	Mean (SD)	65.9 (10.0)	64.9 (11.1)	0.422
Gender	Male	212 (58.7%)	48 (52.2%)	0.257
	Female	149 (41.3%)	44 (47.8%)	
Age at leaving full-time education	16 years or less	271 (75.1%)	73 (79.3%)	0.391
	17 years or more	90 (24.9%)	19 (20.7%)	
Duration of diabetes (months)	Mean (SD)	57.7 (55.9)	63.0 (60.0)	0.420
	Median (range)	36 (1–384)	45 (3–276)	
Therapy	Diet only	98 (27.1%)	26 (28.3%)	0.802
	Monotherapy	136 (37.7%)	37 (40.2%)	
	Combined oral therapy	127 (35.2%)	29 (31.5%)	
Diabetes-related co-morbidity?	Yes	90 (24.9%)	13 (14.1%)	0.026
	No	271 (75.1%)	79 (85.9%)	

* *P* calculated with χ^2 -test for categorical data except for use of Students' *t*-test for age and duration of diabetes.
SD, standard deviation.

by preliminary research in which beliefs were elicited in a series of thirty interviews and incorporated in a questionnaire. Factor analysis of belief statements was then used to identify scales, based on questionnaires completed by 200 primary care-based patients with non-insulin-treated diabetes [16]. Beliefs about medication were explored with a standard questionnaire (Beliefs about Medicines Questionnaire; BMQ) [17]; diabetes self-care activities were explored with a questionnaire including four domains (Summary of Diabetes Self-Care Activities; SDSCA) [18]; and medication adherence was measured with a standard five-item questionnaire (the Medication Adherence Report Scale; MARS) [19]; satisfaction with treatment was measured with a three-domain questionnaire (Diabetes Treatment Satisfaction Questionnaire; DTSQ) [20]; and well-being with the 12-item Well-Being Questionnaire (WBQ-12) [21,22]. Additional demographic and clinical data were collected to characterize the groups.

Procedures

Patients were randomized to their allocated intervention by a study coordinator who had no contact with patients. Baseline questionnaires were completed before randomization. Final questionnaires were sent to patients before the final visit and were completed before seeing the nurse for final assessment. Questionnaire responses were entered onto computer by staff unaware of intervention allocation.

The trial was funded by grants from the National Institutes for Health Research Health Technology Assessment programme who had no role in data analysis, interpretation or decision about publication.

Analysis

Data are presented as means with standard deviations. Internal consistency as assessed by Cronbach's coefficient alpha was calculated in the trial sample for the IPQ-R, BMQ and

internally developed SMBG scales. Characteristics of responders and non-responders were compared with a χ^2 -test statistic for categorical data and Students' *t*-tests for continuous data. Analysis of covariance (ANCOVA) with adjustment for baseline measure was used to compare differences in changes of beliefs between groups. *P*-values were considered significant at 0.05.

Where statistically significant differences were obtained for differences in change of self-reported measures of health behaviour and psychosocial variables between groups, Cohen's *d* standardized effect sizes are reported: by convention, *d* = 0.2 is a small effect, *d* = 0.5 moderate and *d* = 0.8 a large effect [23].

Formal mediation analyses was conducted using standard procedures to establish whether between-group differences over time in self-report outcome measures were mediated by beliefs about diabetes [24]. Unstandardized residuals, saved after the baseline scores were used to predict the follow-up scores, were used as dependent variables to ensure the mediation analyses reflected change in outcome measures.

Results

Of the 453 patients randomized in the trial, 339 (74.8%) completed questionnaires at baseline and 1 year and were included in the final analysis. Figure 1 shows the disposition of trial participants. Table 2 compares the characteristics of responders and non-responders; there were no statistically significant differences between groups apart from an increased proportion of patients with diabetes related co-morbidity among responders compared with non-responders.

Differences between groups in belief changes over time

Group differences between the mean scores on belief measures at follow-up, adjusted for baseline scores, are shown in Table 3. Changes in illness beliefs did not significantly differ

between groups, with the exception of beliefs about consequences ($P = 0.004$). The mean difference in change in consequences scores from baseline to 12 months between the control group and less intensive intervention group (not adjusted for baseline) was 0.92 (95% confidence intervals -0.07 to 1.91 ; Cohen's $d = 0.19$) and between the control group and the more intensive intervention group was 1.59 (0.66 to 2.51; $d = 0.36$).

A significant difference was also found in the change in mean belief scores between the three groups, for beliefs concerning feeling negative about self-testing ($P < 0.001$) and the importance of self-testing ($P < 0.001$). The mean difference in change in feeling negative about self-testing scale scores from baseline to 12 months between the control group and less intensive intervention group (not adjusted for baseline) was -1.37 (-1.83 to -0.91 ; $d = 0.57$) and between the control group and more intensive intervention group was -1.52 (-2.01 to -1.02 ; $d = 0.63$). The mean difference in change in the importance of self-testing scale scores from baseline to 12 months between the control group and less intensive intervention group (not adjusted for baseline) was 0.72 (0.24 to 1.19; $d = 0.31$) and between the control group and more intensive intervention group was 1.08 (0.55 to 1.61; $d = 0.45$). No differences in change between groups were found in either of the other two measures concerning beliefs about SMBG, nor were differences found in changes in beliefs about medication between groups.

Differences between groups in outcome changes over time

Group differences between the mean scores on behavioural and emotional measures at follow-up, adjusted for baseline scores, are shown in Table 4. There were significant differences in scale score changes on the general diet scale of the SDSCA between groups ($P = 0.014$), as well as the specific diet items concerning fruit and vegetables ($P = 0.006$) and high-fat foods ($P = 0.022$). The mean difference in change in SDSCA general diet scores from baseline to 12 months between the control group and less intensive intervention group (not adjusted for baseline) was 0.12 (-0.33 to 0.57 ; $d = 0.06$) and between the control group and more intensive intervention group was -0.50 (-1.00 to 0.01 ; $d = 0.23$). The mean difference in change in the SDSCA fruit and vegetables item from baseline to 12 months between the control group and less intensive intervention group (not adjusted for baseline) was -0.26 (-0.72 to 0.19 ; $d = 0.12$) and between the control group and more intensive intervention group was -0.79 (-1.30 to -0.28 ; $d = 0.34$). The mean difference in change in the SDSCA high-fat foods item from baseline to 12 months between the control group and less intensive intervention group (not adjusted for baseline) was -0.03 (-0.48 to 0.40 ; $d = 0.02$) and between the control group and more intensive intervention group was 0.51 (0.01 to 1.00; $d = 0.23$).

No differences were found for either self-reports of exercise or medication adherence between groups or between groups

over time. Equally, treatment satisfaction or well-being scores did not differ between groups or between groups over time.

Mediation analyses

There were statistically significant differences in mean change scores between groups on three self-report outcome measures: general diet ($P = 0.014$), specific diet (fruit and vegetables, $P = 0.006$), specific diet (high fat, $P = 0.022$). The only measure of beliefs about diabetes that was significantly different between groups was beliefs about consequences ($P = 0.004$). Of the three self-report measures, only change in self-reported fruit and vegetable consumption was predicted by consequences beliefs ($\beta = -0.125$, $P = 0.031$). However, in a regression analysis, with group allocation and beliefs about consequences used to predict change in self-reported fruit and vegetable consumption, group allocation remained a significant predictor ($\beta = -0.168$, $P = 0.004$), whilst beliefs about consequences were non-significant ($\beta = -0.099$, $P = 0.086$). Thus, there was no evidence of even partial mediation of the effects of the intervention on any behavioural measure, by illness beliefs.

Discussion

Among patients allocated to use of more intensive self-monitoring of blood glucose compared with those allocated to usual care over 1 year, there is a small, but significant increase in level of beliefs about the severity of the consequences of diabetes. However, there were no differences between groups in mean changes in beliefs about personal control over diabetes and the perceived effectiveness of treatment. In addition, there was no observed effect on the mean change in well-being between groups. There was also a small but significant effect on self-reported dietary behaviour, with patients allocated to usual care reporting changes towards healthier eating patterns compared with those allocated to more intensive monitoring. Change in beliefs about the consequences of diabetes did not mediate the self-reported changes in dietary behaviour observed between groups.

Strengths and limitations

This trial has provided the first experimental data of the impact of an intervention based on self-monitoring of blood glucose on illness beliefs [10] and self-reported behaviour. It was carried out over 1 year in a group of people broadly representative of patients with Type 2 diabetes [9]. The design addressed the critiques made of earlier self-monitoring studies [7,25] and used a standardized programme to teach patients active self-management using the information acquired from self-monitoring [11]. In addition, the experimental design employed allowed the examination of causal relations between beliefs and behaviour, and not just associations, which may be affected by confounding. The use of a widely accepted psychological model on which to base design of intervention

Table 3 Comparison of changes in mean scores (with standard deviations) between groups on measures assessing beliefs about illness, self-monitoring of blood glucose and medication from baseline to 1 year

		Usual care: no meter <i>n</i> = 113				Self-testing <i>n</i> = 121				Self-monitoring <i>n</i> = 105				
	Min–max range	Baseline		Follow-up		Baseline		Follow-up		Baseline		Follow-up		<i>P</i> value for difference
Illness perceptions (IPQ-R)														
Identity*	0 to 14	2.2	(2.8)	1.7	(2.2)	1.7	(2.2)	2.0	(2.4)	1.8	(2.1)	1.8	(2.0)	0.054
Timeline—acute/chronic†	6 to 30	24.5	(4.1)	25.0	(3.6)	24.4	(4.0)	24.6	(4.3)	25.3	(3.5)	25.4	(4.0)	0.61
Timeline—cyclical‡	4 to 20	10.4	(3.3)	10.0	(3.1)	10.4	(3.0)	10.4	(3.1)	10.6	(2.9)	10.7	(3.1)	0.33
Consequences§	6 to 30	17.3	(3.9)	16.8	(4.3)	16.9	(3.7)	17.4	(4.2)	17.2	(3.4)	18.3	(3.8)	0.004
Personal control¶	6 to 30	24.1	(3.1)	24.3	(2.8)	24.2	(2.7)	24.1	(3.5)	24.7	(3.1)	25.0	(3.6)	0.30
Treatment control**	5 to 25	18.2	(1.9)	17.8	(2.0)	18.3	(2.0)	18.3	(1.9)	18.6	(2.0)	18.1	(2.5)	0.29
Illness coherence††	5 to 25	16.9	(4.1)	17.4	(4.4)	16.6	(4.5)	17.8	(4.1)	17.1	(4.1)	18.2	(4.1)	0.36
Emotional representations‡‡	6 to 30	15.3	(4.3)	15.0	(4.0)	15.1	(4.7)	15.3	(4.6)	15.6	(4.7)	15.4	(4.8)	0.58
Self-monitoring blood glucose beliefs														
Feel negative about self-testing§§	2 to 10	5.5	(1.8)	5.9	(1.9)	5.5	(1.7)	4.6	(1.4)	5.3	(1.7)	4.3	(1.6)	< 0.001
Symptoms make test unnecessary¶¶	2 to 10	5.1	(1.4)	5.2	(1.7)	5.0	(1.6)	5.3	(1.7)	5.1	(1.8)	5.2	(1.7)	0.96
Important to self-test***	2 to 10	6.4	(1.4)	6.1	(1.3)	6.4	(1.4)	6.9	(1.5)	6.5	(1.6)	7.3	(1.6)	< 0.001
Checking is painful†††	1 to 5	2.1	(0.8)	2.2	(0.9)	2.3	(0.8)	2.1	(0.9)	2.2	(0.8)	2.0	(0.9)	0.21
Beliefs about medication (BMQ)														
Necessity‡‡‡	5 to 25	19.1	(2.3)	18.9	(2.4)	18.5	(3.2)	18.6	(2.9)	18.7	(2.9)	19.4	(2.8)	0.28
Concerns§§§	5 to 25	13.7	(2.9)	13.3	(3.2)	13.6	(3.9)	13.1	(3.5)	13.3	(3.8)	13.4	(3.9)	0.75

Missing data for scales: *0; †25; ‡23; §25; ¶19; **26; ††29; ‡‡18; §§31; ¶¶40; ***37; †††35; ‡‡‡104; §§§106.

BMQ scales were only completed by patients taking medication at baseline.

BMQ, Beliefs about Medicines Questionnaire; IPQ-R, revised Illness Perceptions Questionnaire.

Table 4 Comparison of changes in mean scores (with standard deviations) between groups on behavioural and emotional outcome measures from baseline to one year

		Usual care: no meter <i>n</i> = 113				Self-testing <i>n</i> = 121				Self-monitoring <i>n</i> = 105				
	Min-max range	Baseline		Follow-up		Baseline		Follow-up		Baseline		Follow-up		<i>P</i>
Self-care behaviours (SDSCA)														
General diet*	0 to 7	5.2	(1.8)	5.6	(1.5)	5.0	(1.8)	5.5	(1.3)	5.2	(1.8)	5.1	(1.9)	0.014
Specific diet†	0 to 7	5.4	(1.2)	5.9	(1.1)	5.3	(1.2)	5.7	(1.0)	5.5	(1.4)	5.4	(1.3)	< 0.001
Specific diet: fruit and vegetables‡	0 to 7	5.2	(1.8)	5.7	(1.6)	5.1	(2.0)	5.4	(1.7)	5.4	(1.9)	5.2	(2.0)	0.006
Specific diet: high-fat foods§	0 to 7	2.4	(1.7)	1.9	(1.5)	2.5	(1.6)	2.0	(1.4)	2.3	(1.8)	2.4	(1.5)	0.022
Exercise¶	0 to 7	3.3	(2.1)	4.0	(2.2)	3.2	(2.2)	3.6	(2.2)	3.4	(2.2)	3.9	(2.4)	0.45
Medication adherence (MARS)**	5 to 25	24.0	(1.6)	24.1	(2.0)	23.9	(1.6)	24.0	(1.4)	24.0	(1.3)	24.1	(1.2)	0.91
Diabetes Treatment Satisfaction (DTSQ)														
Total satisfaction††	0 to 36	29.3	(6.8)	30.0	(5.3)	29.4	(6.5)	29.7	(5.6)	29.7	(5.4)	30.1	(5.5)	0.93
Perceived frequency of hyperglycaemia‡‡	0 to 6	1.7	(1.7)	1.9	(1.9)	1.5	(1.6)	2.3	(1.5)	2.0	(1.7)	2.4	(1.7)	0.053
Perceived frequency of hypoglycaemia§§	0 to 6	0.6	(1.2)	0.7	(1.3)	0.7	(1.3)	0.7	(1.2)	0.8	(1.3)	0.8	(1.3)	0.97
WBQ-12														
Total well-being¶¶	0 to 36	25.1	(6.3)	25.9	(5.8)	24.3	(6.8)	24.5	(7.0)	25.2	(6.3)	24.9	(6.4)	0.38
Negative well-being***	0 to 12	1.5	(2.1)	1.3	(2.0)	1.6	(2.3)	1.4	(2.2)	1.4	(2.2)	1.4	(2.2)	0.92
Energy†††	0 to 12	6.5	(2.9)	6.8	(2.6)	6.3	(2.9)	6.4	(2.9)	6.6	(2.8)	6.7	(2.7)	0.73
Positive well-being‡‡‡	0 to 12	7.6	(3.2)	8.3	(2.8)	7.6	(3.2)	7.6	(3.4)	7.9	(3.0)	7.6	(3.0)	0.20

Missing data for scales: *40; †32; ‡28; §30; ¶37; **96; ††11; ‡‡92; §§93; ¶¶32; ***18; †††25; ‡‡‡19.

MARS only completed by those taking medication at baseline.

MARS, Medication Adherence Report Scale; SDSCA, Summary of Diabetes Self-Care Activities; WBQ, Well-Being Questionnaire.

and assessment enables comparisons to be made with an extensive body of previous non-experimental research.

The limited response to questionnaires raises the possibility of systematic differences between responders and non-responders. However, there is little evidence of this on the basis of demographic characteristics. The exclusion of patients who tested regularly may have resulted in the trial excluding those patients who found SMBG to be of most value. The target group in this study comprised patients with HbA_{1c} values that were open to improvement and who might benefit from SMBG to achieve this improvement. However, the recent finding of similar effects of SMBG on HbA_{1c} to this study in the ESMON trial of newly diagnosed patients with Type 2 diabetes [26] suggests that the findings may apply more widely across patient groups.

The self-report questionnaires used to measure changes in behaviour may not have been as responsive to change as ideal. Although the summary of diabetes self-care activities scale is the most widely used measure of diabetes self-management, test-retest correlations are often moderate and associations with other measures of behaviour are often small [18].

Strengths and limitations in comparison with other studies

Previous studies have suggested that patients exposed to use of a meter might achieve an increased sense of personal control over their condition as a result of seeing the impact of their behaviour on blood glucose levels. [5,6] However, observational studies do not allow the impact of using a meter to be separated from other support for self-management. The use of self-management strategies and goal setting for all groups in the present trial allowed an appropriate comparison group to be maintained to assess the impact of SMBG in conjunction with self-management compared with self-management support alone on beliefs and behaviour.

Concerns have been raised that self-monitoring may have negative effects and in this study we demonstrated that SMBG led to stronger beliefs that diabetes has negative consequences in the self-monitoring groups. In an in-depth interview, study patients reported that when the blood glucose readings did not make sense it led to frustration [5]. Equally, the recent ESMON trial found that the regular use of SMBG in patients newly diagnosed with Type 2 diabetes lead to increased reports of depression [26]. We have previously reported a small decrease in health status measured with the EuroQol 5-D over 1 year among patients in the trial allocated to more intensive monitoring compared with usual care [27]. However, this is not confirmed by the results of the widely used and validated WBQ, which indicate that well-being remains moderately good or stable over the trial period, with no significant difference between groups, and treatment satisfaction remains high.

Previous studies have shown that illness beliefs are associated with adherence to self-care behaviour such as diet and physical activity in patients with Type 2 diabetes [28–32].

However, findings of associations between beliefs about treatment effectiveness and self-monitoring have not been consistent. One found an association, [31] whilst others did not [29,30]. We observed no difference in mean changes in beliefs about treatment effectiveness between allocated groups, although beliefs about self-monitoring were significantly less negative in patients allocated to use of self-monitoring compared with those allocated to usual care.

The present study found that patients in the control group reported significant improvements in their diet, relative to the more-intensive self-monitoring group, with the less intensive group reporting non-significant improvements. These changes were in terms of general diet, increased fruit and vegetable consumption and reduced fat consumption. These findings could either represent a true failure to change diet in those using SMBG, perhaps because of attention to glucose readings rather than dietary change or, alternatively, reflect an increased knowledge in patients engaging in SMBG of the deficiencies of their diet.

Meaning

The broadly positive assessment of use of blood glucose self-monitoring in a recent systematic review [7] is largely based on the common-sense approach that, given information about blood glucose levels and sufficient support, patients will be able to make the necessary changes to their lifestyle. Despite a careful critique of previous trials, the review does not draw a possible conclusion that limitations in design, conduct and analysis of these trials may have led to an overestimation of the extent of the mean effect of self-monitoring among non-insulin-using patients with Type 2 diabetes. In our trial, we assessed different methods of glycaemic feedback and use of SMBG does not appear to offer more motivation than use of HbA_{1c}.

Use of SMBG to provide feedback about the impact of brisk walking or of a meal high in refined sugar is likely to provide limited information to some groups of patients. However, the extent to which personal experimentation, in addition to being taught about the importance of these health-related behaviours, has an effect on long-term adherence to diet is not clear. It is possible that, for many patients, the mean day-to-day variability of self-monitored blood glucose levels may mask the small benefits that accrue from small, but important, changes in health behaviours over the same period. Less variable measures such as HbA_{1c} repeated at longer intervals, may identify changes in glycaemic control with more precision than self-monitoring and provide positive feedback about the health gains from small and sustainable changes in diet and physical activity.

Future work

Despite extensive previous efforts to assess the effectiveness of self-monitoring of blood glucose for non-insulin-treated

patients with Type 2 diabetes, this is the first trial in which potential mediators of effect have been explored. Observed differences in mean changes in beliefs about diabetes between patients allocated to self-monitoring and usual care are consistent with the lack of convincing evidence for a clinically significant effect on glycaemic control. Further experimental work is required to explore the utility of the CSM in explaining changes in health-related behaviours. In addition, there may be groups of Type 2 diabetes patients where self-monitoring may be particularly useful; for example, those treated with insulin. Understanding the impact of self-monitoring in these circumstances may enable more effective use and better design of patient support programmes.

Competing interests

Nothing to declare.

Acknowledgements

A full list of acknowledgements and membership of the DiGEM trial group can be found elsewhere [9]. The views and opinions expressed in this report are not necessarily those of the Department of Health. The trial was funded by the NIHR Health Technology Assessment Programme and the NHS. Dr Farmer, Professor Kinmonth and Professor Neil are members of the NIHR School of Primary Care Research.

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